

TARGET SHIP DATE: 2026-04-22
Gerrymandering Simulator by shiii_man

Roster: Isabel Zheng (PM), Haowen Xiao, Sean Takahashi, Araf Hoque, and Owen Zeng

Overview:

Using the New York Open Data files on voter registration by NYC election districts and the median salaries by congressional district, we can identify whether district boundaries divide populations in ways consistent with gerrymandering (ex. Packing wealthy voters together). We intend for this to be interactive, illustrating a map of New York (colors indicate varying levels of income & drawn county and congressional district lines) and allowing users to learn how gerrymandering is done. This was our inspiration for this project in terms of detailed voting results: <https://votehub.com/2024-map/>

Database Organization:

profiles		
INTEGER	id	PK AUTOINCREMENT
TEXT	username	UNIQUE
TEXT	password	
INTEGER	total_maps	
INTEGER	wins	

All gerrymandered maps

saved_maps		
INTEGER	id	PK AUTOINCREMENT
INTEGER	user_id	
TEXT	map_name	UNIQUE
TIMESTAMP	created_at	CURRENT_TIMESTAMP
TEXT	player_party	“DEM” or “REP”
INTEGER	District1 – district20	JSON arrays of election IDs (one column per assembly district)

Core Components:

1. CSV Dataset Files
 - a. Both datasets are relational. To observe the potential gerrymandering techniques, the data needs to be stored in rows and columns; the overlaps between voting districts and ZIP codes in terms of their geographical area.
 - i. Voter Data 2008-2018
(https://data.cityofnewyork.us/City-Government/Voter-Analysis-2008-2018/psx2-aqx3/about_data): Gives data of NYC voters who have voted between 2008 and 2018 including their party registration, what elections they voted in, and their geographic location(voting district)
 - ii. Salary Information 2018-Adjusted
([https://data.census.gov/all?q=median+salary&g=040XX00US36\\$8600000](https://data.census.gov/all?q=median+salary&g=040XX00US36$8600000)): Taken from Census data, gives salary information including median, mean, quartile of yearly income by ZIP code.
2. GeoJSON files
 - a. nyled_18d.geojson: Each election district stored as their geographic area
 - b. ZipCode.geojson (renamed from ny_zipcodes.geojson): NYC based on ZIP code.
3. Python
 - a. __init__.py: Serve the Flask app, base file
 - b. auth.py: Basic login/register functionality
 - c. [utility.py](#): Database helper functions (dictify, insert_query, general_query, get_user)
 - d. Build_db.py: Script that drops/recreates all tables and loads CSV/GeoJSON data into SQLite
4. Folium for Visualization (used for the static income/voter explorer map on the home page only; generates JS and HTML with location points)

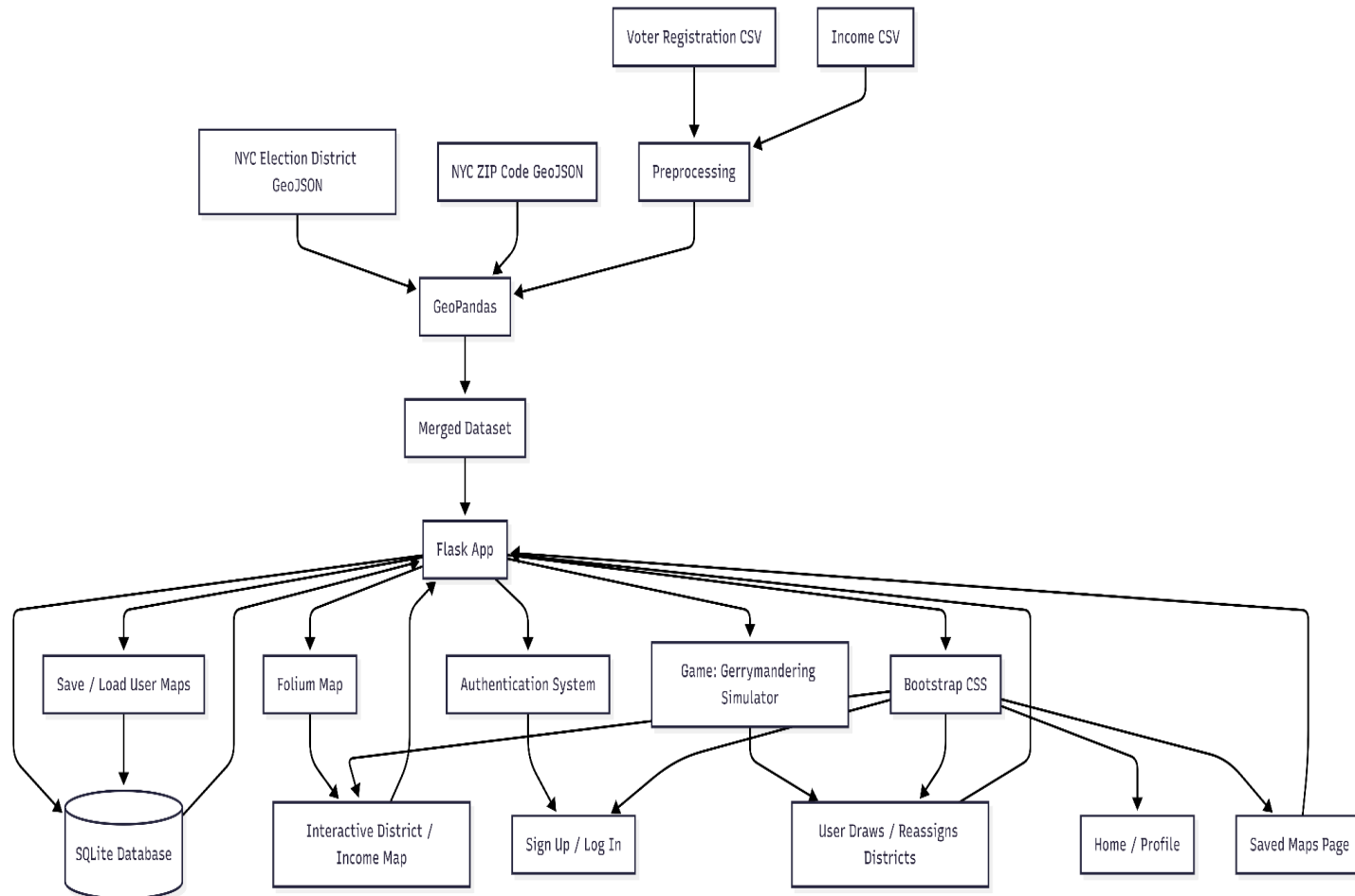
Front-end Framework: Bootstrap (contains prebuilt CSS classes for buttons, containers, grid layouts. These prebuilt classes make styling much easier and more time efficient, allowing us to focus more on the backend: game logic, data processing, etc)

APIs: Leaflet.js is loaded via CDN (no server-side API key required). CartoDB provides the tile layer. No additional external APIs are used.

Data Visualization Library:

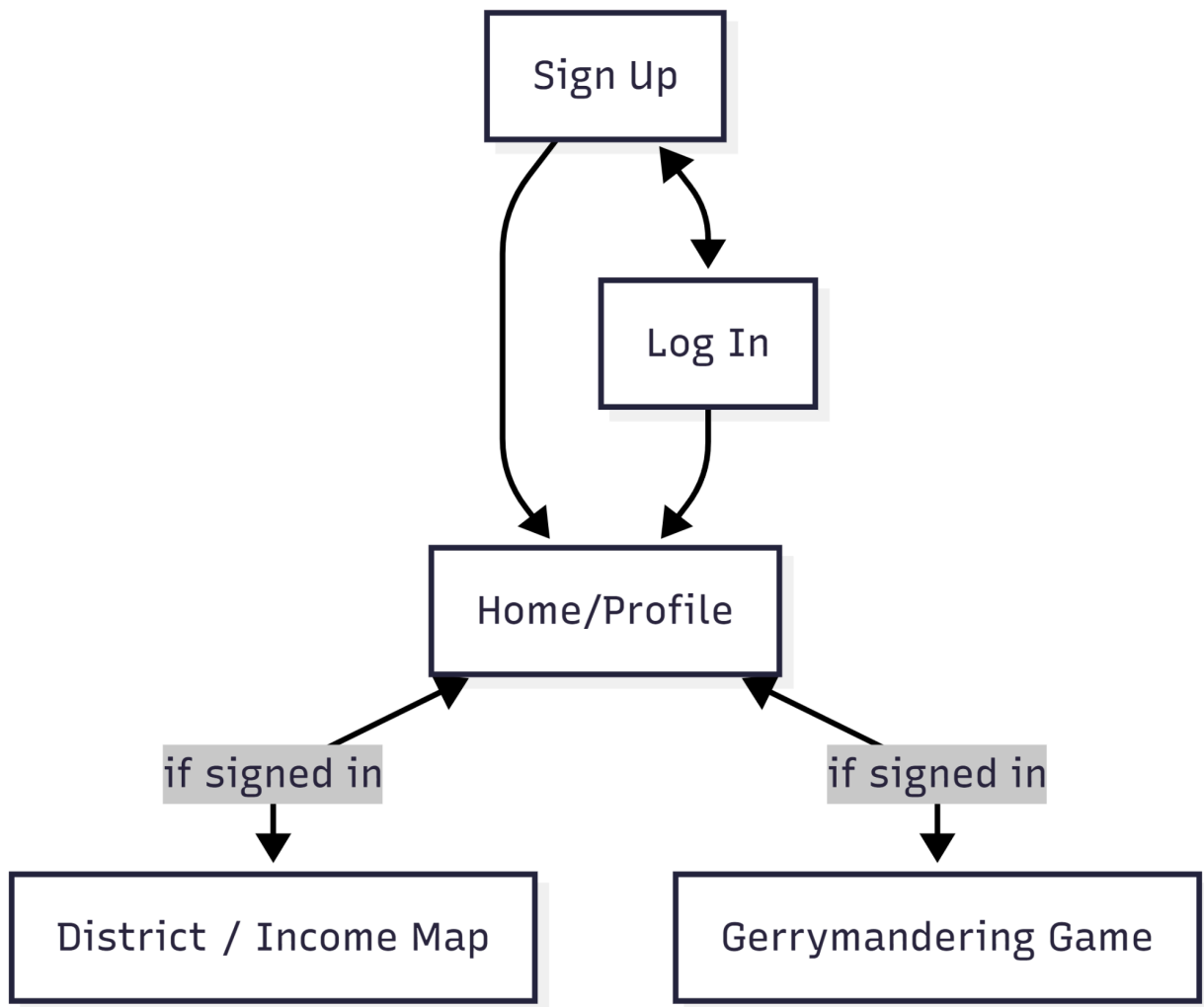
- Folium: Used to generate the static income/voter explorer map on the home page. Takes spatial and relational data to show election districts, ZIP codes, income distribution, and voter distribution with colors and lines.
- [Leaflet.js](#): Client-sided interactive map library used for the gerrymandering game. Used with CDN with CartoDB. Displays all NYC election districts with color-coded status (unassigned, in-progress, neighbor, finalized) and supports interactive district selection.

Component Map:



[\(Mermaid Live Editor\)](#)

Site Map:



([Mermaid Live Editor](#))

<https://gist.github.com/abatko/ee7b24db82a6f50cfce02afafa1dfd1e>